

5. State Pascal's law.

Ans. Pascal's law states that the external pressure applied on an incompressible liquid is transmitted uniformly throughout the liquid.

[QY-'19 & '23; HY-'23; April-'24 & '25]

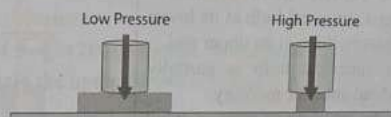
VI. Answer in detail.

(SLOW LEARNERS' ANSWERS GIVEN AT THE END OF THE UNIT)

1. With an appropriate illustration prove that the force acting on a smaller area exerts a greater pressure.

- Ans. (i) Take a nail, which has two ends: One end is sharp and other end is a bulged head.
 (ii) We usually keep the pointed end on the wall or wood and hammer on the bulged head.
 (iii) The force applied on the sharp end acts over a very small area, creating a large pressure.
 (iv) Thus the nail penetrates into the wall or wood.
 (v) Thus the force acting on a smaller area exerts a greater pressure.

$$\text{Pressure} = \frac{\text{Thrust}}{\text{Area of contact}}$$



2. Describe the construction and working of mercury barometer.

[HY-'23]

Ans.

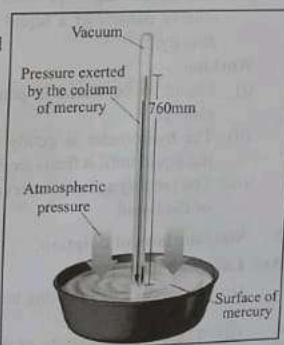
- (i) **Mercury Barometer** is the instrument used to measure atmospheric pressure.
 (ii) It is designed by Torricelli.

Construction :

- (i) Mercury Barometer consists of long glass tube, closed at one end and opened at the other.
 (ii) Mercury filled through open end and close that end by thumb and open it after immersing it into a trough of mercury.

Working :

- (i) The Barometer works by balancing the mercury in the glass tube against the outside air pressure.
 (ii) If the air pressure increases, it pushes more of the mercury up into the tube.
 (iii) If the air pressure decreases, more of the mercury drains from the tube.



III. State whether true or false. If false, correct the statement.

1. The weight of fluid displaced determines the buoyant force on an object.
Ans. True.

2. The shape of an object helps to determine whether the object will float or not.

Ans. False. Correct statement : The **density** of an object helps to determine whether the object will float or sink.

3. The foundations of high-rise buildings are kept wide so that they may exert more pressure on the ground.

Ans. False. Correct statement : The foundations of high-rise buildings are kept wide so that they may exert less pressure on the ground.

4. Archimedes' principle can also be applied to gases.

Ans. True.

5. Hydraulic press is used in the extraction of oil from oil seeds.

Ans. True.

IV. Match the following.

Column I	Column II	Answer
Density	$h\rho g$	Mass Volume
Pascal's law	Milk	Pressure
Pressure exerted by a fluid	Mass Volume	$h\rho g$
Lactometer	Pressure	Milk

V. Answer in brief.

(SLOW LEARNERS' ANSWERS GIVEN AT THE END OF THE UNIT)

1. On what factors the pressure exerted by the liquid depends on? [QY-'23]

Ans. The pressure exerted by the liquid depends on the

- (i) Depth of the liquid
- (ii) Density of the liquid
- (iii) Acceleration due to gravity.

2. Why does a helium balloon float in air?

Ans. Helium balloon floats in air because helium gas is less dense than air. [QY-'19; HY & April-'23]

3. Why it is easy to swim in sea water than in river water? [QY-'19]

Ans. The reason is that the sea water contains salt and so its density is more than the density of river water. Sea water provides stronger buoyant force. The weight of a man gets balanced by the less immersed part of his body in sea water as compared to river water. Thus, it is easier for a person to swim in sea water than in river water.

4. What is meant by atmospheric pressure?

Ans. Earth is surrounded by a layer of air up to certain height (nearly 300 km) and this layer of air around the earth is called atmosphere of the earth. Since air occupies space and has weight, it also exerts pressure. This pressure is called atmospheric pressure. [HY-'19; April-'23]

5. State Pascal's law.

Ans. Pascal's law states that the pressure applied to a fluid is transmitted equally in all directions throughout the fluid.

1. With an application, state Pascal's law.

Ans. (i) Take a glass bottle.

(ii) We usually use a bottle to store liquid.

(iii) The force applied to the liquid is transmitted equally in all directions.

(iv) Thus the pressure is transmitted equally in all directions.

(v) Thus the pressure is transmitted equally in all directions.

Pressure

2. Describe the construction of a barometer.

Ans.

(i) Mercury is used to measure the pressure.

(ii) It is designed to measure the pressure.

Construction :

(i) Mercury is used to measure the pressure.

(ii) Mercury is used to measure the pressure.

(iii) Mercury is used to measure the pressure.

Working :

(i) The Barometer is used to measure the pressure.

(ii) If the air pressure is high, the mercury level in the tube rises.

(iii) If the air pressure is low, the mercury level in the tube falls.